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# EXPERIMENT 5

**Title:**Automating Application Deployment using GitHub and Jenkins

**Aim:**To automate the application build, testing, and deployment process using GitHub and Jenkins.

**Objectives**

- To create a GitHub repository.
- To create and maintain source code using GitHub.
- To install and configure Jenkins.
- To integrate Jenkins with GitHub.
- To create a Jenkins Pipeline.
- To automate Build, Test, and Deploy stages.
- To verify successful pipeline execution.

**Requirements:**

Hardware

- Computer/Laptop
- Internet connection

Software

- Windows OS
- Git
- Java JDK 21
- Jenkins
- GitHub
- Web browser

**Procedure**

**Step 1**: Create GitHub Repository

Create a GitHub repository named:
jenkins-demo
Initialize the repository with a README file.

**Step 2:** Create index.html

Create a file named:
index.html
Add the following code:

```
<!DOCTYPE html>
<html>
```

```
<head>
   <title>Jenkins Demo</title>
</head>
<body>
   <h1>Hello Jenkins!</h1>
</body>
</html>
```

Commit the file to the main branch.

**Step 3:** Install Java

Install Java JDK 21.

Verify the installation using:

java -version

Java 21 should be displayed.

**Step 4:** Install Jenkins

Install Jenkins on Windows.

During installation, select the Java 21 installation:

C:\Program Files\Eclipse Adoptium\jdk-21.0.12.101-hotspot\

Run Jenkins as a Windows service.

Use port:

8080

**Step 5:** Open Jenkins

Open a web browser and enter:

http://localhost:8080

Unlock Jenkins using the initial administrator password.

**Step 6:** Install Plugins

Select:

Install suggested plugins

Wait until the plugin installation is completed.

**Step 7:** Create Administrator Account

Create a Jenkins administrator account by providing:

Username
Password
Full Name
Email Address

Then open the Jenkins dashboard.

**Step 8:** Create a Pipeline Project

From the Jenkins dashboard, select:

New Item → jenkins-demo → Pipeline → OK

**Step 9:** Create Pipeline Script

Under Pipeline, select:

Pipeline script

Enter the following script:

```
pipeline {
   agent any

   stages {
      stage('Build') {
         steps {
            echo 'Building application...'
         }
      }

      stage('Test') {
         steps {
            echo 'Testing application...'
         }
      }

      stage('Deploy') {
         steps {
            echo 'Deploying application...'
         }
      }
   }
}
```

Click Save.

**Step 10:** Run the Pipeline

Click:

Build Now

Open the generated build number and select:

Console Output

**Step 11:** Create Jenkinsfile

In the GitHub repository, create a file named:

Jenkinsfile

Add:

```
pipeline {
   agent any

   stages {
      stage('Build') {
         steps {
            echo 'Building application...'
         }
      }

      stage('Test') {
         steps {
            echo 'Testing application...'
         }
      }

      stage('Deploy') {
         steps {
            echo 'Deploying application...'
         }
      }
   }
}
```

Commit the file.

The repository should contain:

README.md
index.html

Jenkinsfile

**Step 12**: Connect Jenkins with GitHub

Open:

Jenkins → jenkins-demo → Configure → Pipeline

Change Definition to:

Pipeline script from SCM

Select:

SCM: Git

Repository URL:

https://github.com/gunjanrupani/jenkins-demo.git

Branch:

*/main

Script Path:

Jenkinsfile

Click Save.

**Step 13:** Execute the GitHub Pipeline

Click:

Build Now

Jenkins obtains the Jenkinsfile from GitHub and checks out the latest source code.

The pipeline executes:

Build
↓
Test
↓
Deploy

**Step 14:** Verify Output

Open:

Build Number → Console Output

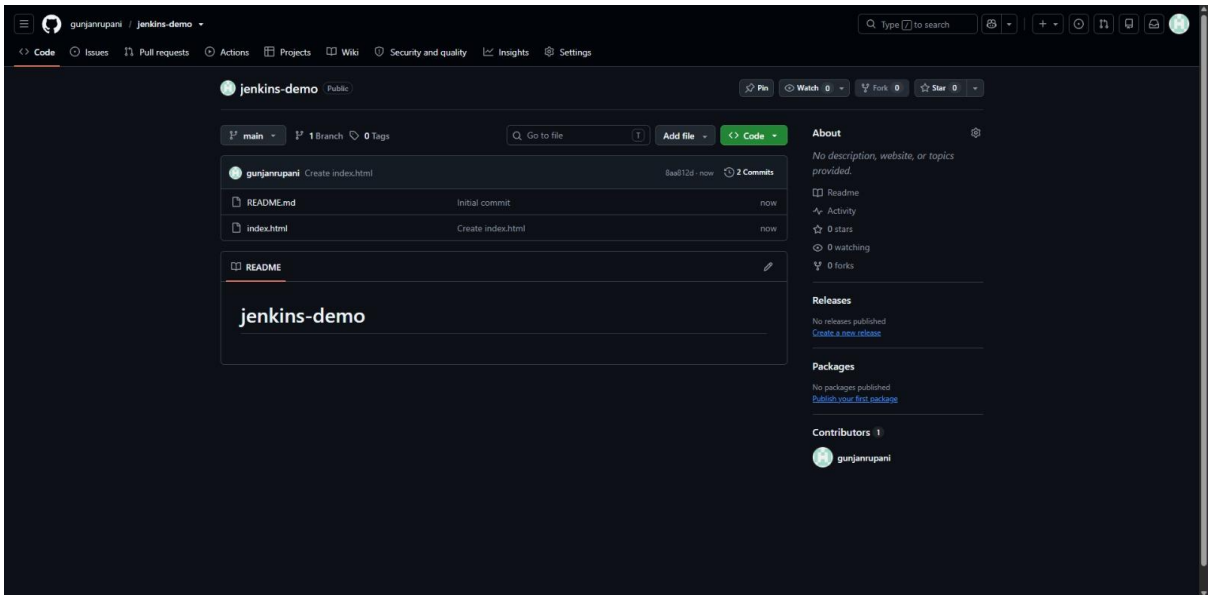
**Step 15:** Modify the Application

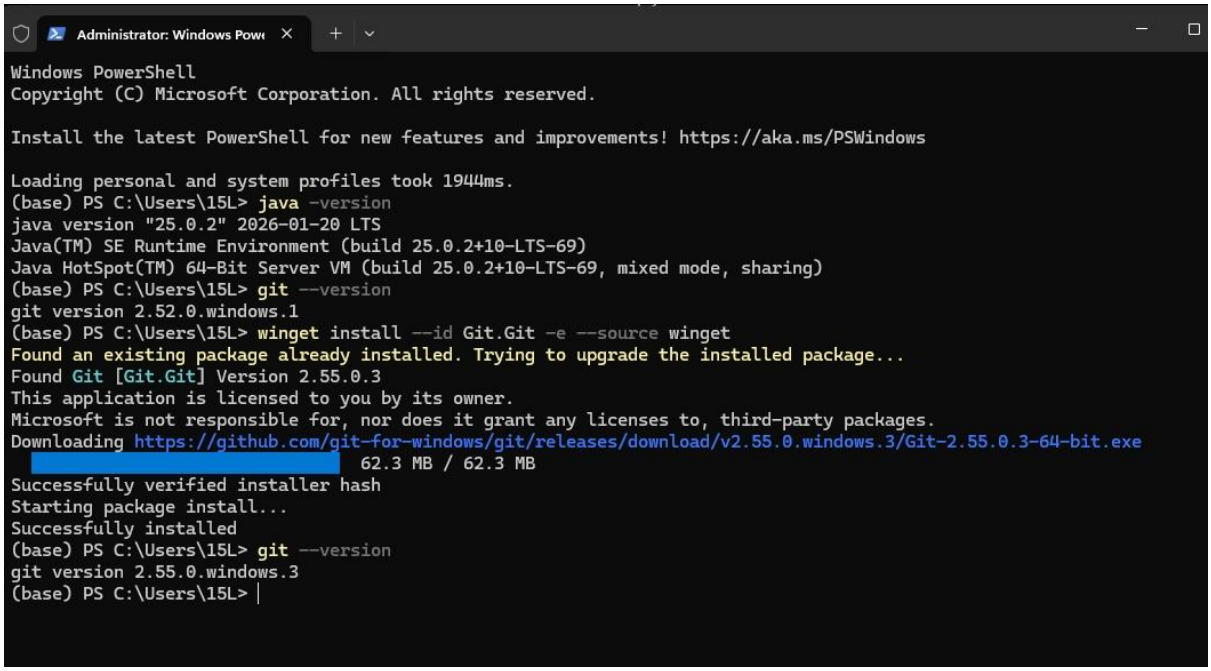
Update index.html in GitHub:

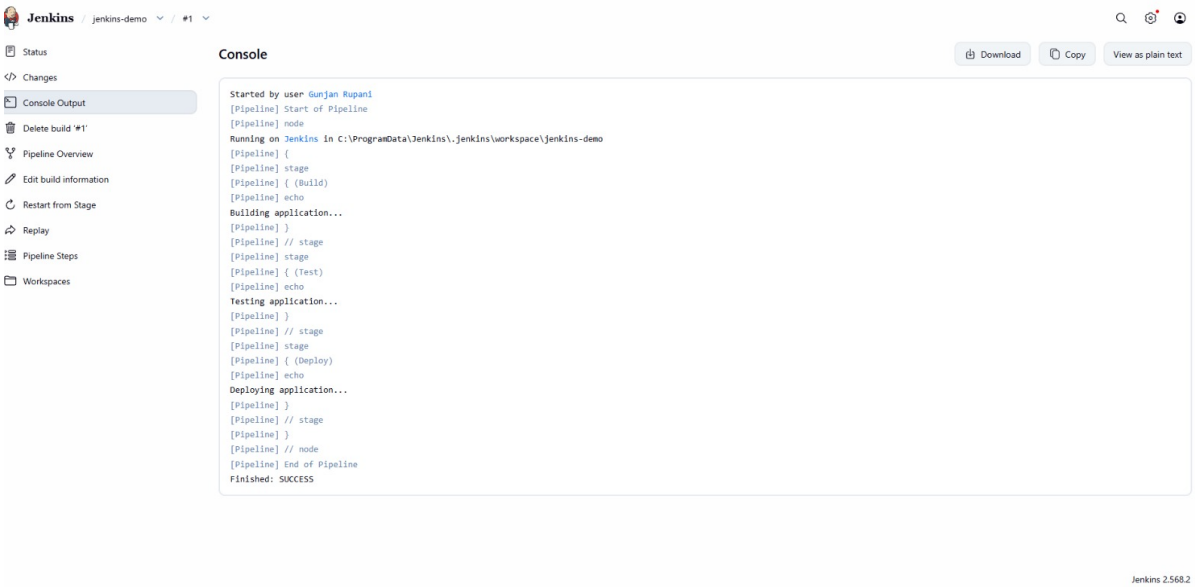
<h1>Hello Jenkins! Version 2</h1>

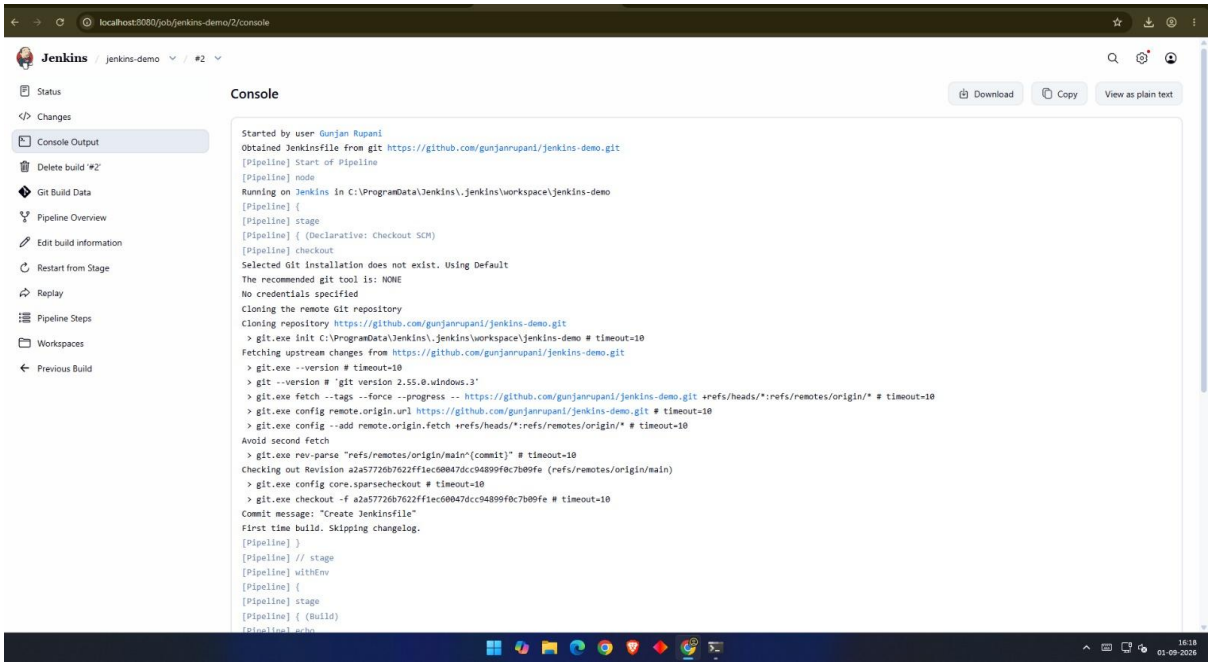
Commit the change.
Run Build Now again in Jenkins.
Jenkins retrieves the updated commit and executes the Pipeline.

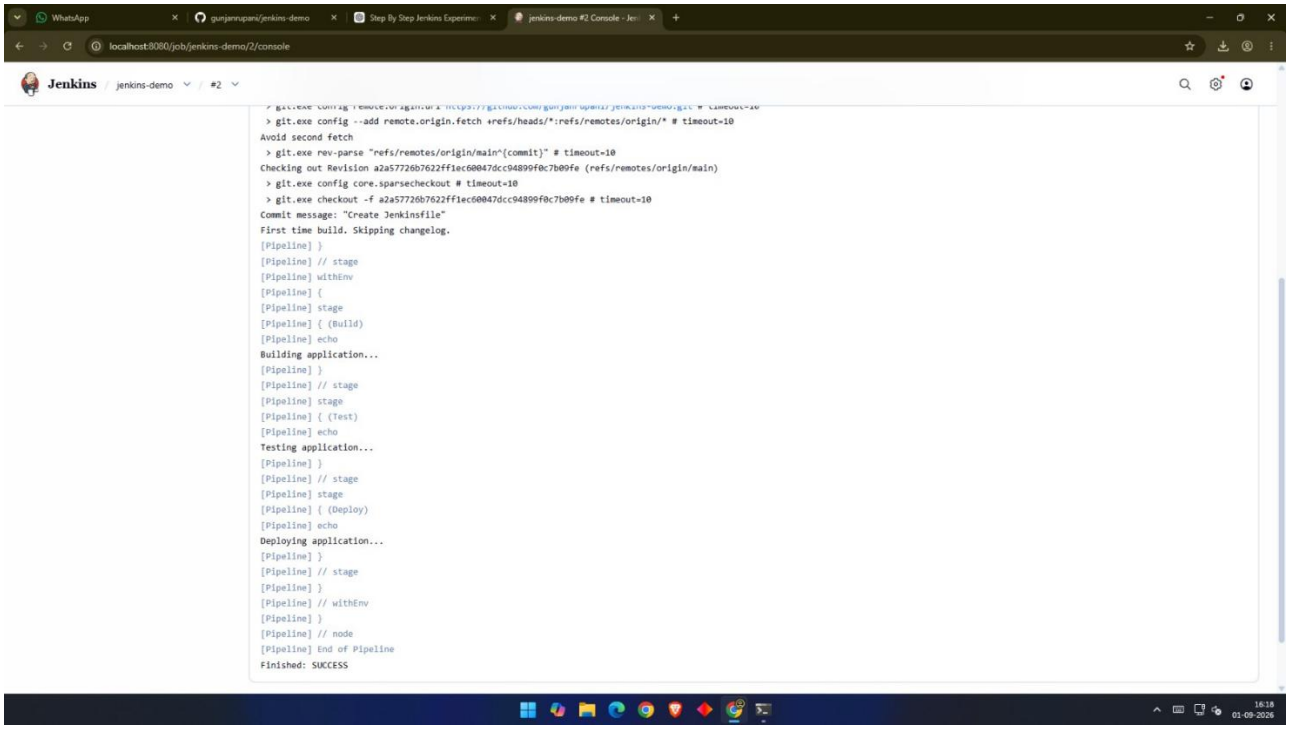
**Output:**




Administrator: Windows Powe
Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.
Install the latest PowerShell for new features and improvements! https://aka.ms/PSWindows
Loading personal and system profiles took 1944ms.
(base) PS C:\Users\15L> java -version
java version "25.0.2" 2026-01-20 LTS
Java(TM) SE Runtime Environment (build 25.0.2+10-LTS-69)
Java HotSpot(TM) 64-Bit Server VM (build 25.0.2+10-LTS-69, mixed mode, sharing)
(base) PS C:\Users\15L> git --version
git version 2.52.0.windows.1
(base) PS C:\Users\15L> winget install --id Git.Git -e --source winget
Found an existing package already installed. Trying to upgrade the installed package...
Found Git [Git.Git] Version 2.55.0.3
This application is licensed to you by its owner.
Microsoft is not responsible for, nor does it grant any licenses to, third-party packages.
Downloading https://github.com/git-for-windows/git/releases/download/v2.55.0.windows.3/Git-2.55.0.3-64-bit.exe
62.3 MB / 62.3 MB
Successfully verified installer hash
Starting package install...
Successfully installed
(base) PS C:\Users\15L> git --version
git version 2.55.0.windows.3
(base) PS C:\Users\15L>


Jenkins
jenkins-demo
#1
Status
Changes
Console Output
Delete build '#1'
Pipeline Overview
Edit build information
Restart from Stage
Replay
Pipeline Steps
Workspaces
Console
Download
Copy
View as plain text
Started by user Gunjan Rupani
[Pipeline] Start of Pipeline
[Pipeline] node
Running on Jenkins in C:\ProgramData\Jenkins\.jenkins\workspace\jenkins-demo
[Pipeline] {
[Pipeline] stage
[Pipeline] { (Build)
[Pipeline] echo
Building application...
[Pipeline] }
[Pipeline] // stage
[Pipeline] stage
[Pipeline] { (Test)
[Pipeline] echo
Testing application...
[Pipeline] }
[Pipeline] // stage
[Pipeline] stage
[Pipeline] { (Deploy)
[Pipeline] echo
Deploying application...
[Pipeline] }
[Pipeline] // stage
[Pipeline] }
[Pipeline] // node
[Pipeline] End of Pipeline
Finished: SUCCESS
Jenkins 2.568.2




**Result**

The GitHub repository was successfully integrated with Jenkins. Jenkins successfully retrieved the Jenkinsfile from GitHub and executed the Build, Test, and Deploy stages. The Pipeline completed successfully with:

Finished: SUCCESS

**Conclusion**

Thus, the application build, testing, and deployment process was successfully automated using GitHub and Jenkins Pipeline.